Vineel Koushik Peruri

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GitHub — LinkedIn

Education

S.R.K Institute of Technology, Vijayawada 2022 – 2026
B.Tech in Computer Science and Engineering (Data Science)
CGPA: 8.35
Final Year Project: Dual-Stage Multimodal Voice Pathology Detection
Conv-HSA + MObjGB-Ensemble — MSRSpT-152

Technical Skills

Languages: Python, Java, JavaScript
Web Development: HTML, CSS, Node.js, Express.js, Flask
Data Science & ML: Pandas, NumPy, Scikit-learn, Matplotlib
AI/LLM Tools: LangChain, Pinecone, Hugging Face Transformers, Groq
Databases: MySQL
Core Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks

Experience

Data Science Intern, BIST Technologies May 2025 – July 2025

- Developed a Road Accident Analysis system using real-world datasets to identify high-risk zones and patterns
- Built ML models for classification and regression to predict accident severity
- Performed data preprocessing and exploratory data analysis on large datasets
- Improved data insights through effective visualization techniques

Projects

Medical AI Assistant (RAG Chatbot)
Tech: *Python, Flask, Pinecone, LangChain, Hugging Face Transformers, Groq*

- Developed a Retrieval-Augmented Generation (RAG)-based medical assistant for answering disease-related queries
- Built embedding-based vector search pipeline for efficient semantic context retrieval
- Integrated 1000+ medical documents (PDFs and Wikipedia) to expand domain knowledge
- Implemented query intent classification (definition, treatment, detailed response)
- Designed structured outputs including Symptoms, Causes, Treatment, and Risk Level
- Optimized response generation pipeline for faster and more relevant outputs

Vendor Invoice Intelligence Dashboard
Tech: *Python, Pandas, NumPy, Streamlit, Matplotlib, Machine Learning*

- Built a Streamlit-based interactive dashboard to analyze vendor invoice data and generate actionable business insights
- Automated duplicate invoice detection and payment delay analysis, reducing manual effort by ∼20–30%
- Applied data preprocessing, feature engineering, and ML models (classification & regression)
- Developed KPI-based visualizations for vendor performance and payment cycles
- Identified patterns in invoice processing to support data-driven financial decisions

Problem Solving

- Solved 120+ problems on LeetCode focusing on core DSA patterns
- Practiced arrays, recursion, binary search, and dynamic programming
- Focused on improving problem-solving with time and space optimization